

Krutik Tikam

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🐙 GitHub | 🔗 LinkedIn

PROFESSIONAL SUMMARY

Versatile AI Software Engineer with end-to-end expertise in building, optimizing, and deploying generative AI pipelines, computer vision tracking systems, and local RAG architectures. Proficient in PyTorch, FastAPI, and agentic frameworks, with a strong track record of engineering low-latency, scalable AI solutions. Adept at utilizing modern developer tools like Claude and Cursor to accelerate system design and deploy high-performance intelligence platforms.

TECHNICAL SKILLS

AI & Machine Learning: PyTorch, YOLO, Computer Vision, RAG Architectures, Agentic Systems, Vector Search (ChromaDB), Deep Learning
Systems & Backend: Python, FastAPI (Async), Rust (Tauri), PostgreSQL, SQLite, RESTful APIs, System Architecture
DevOps & Tooling: Docker, Git/GitHub Actions, Cursor, Claude, AWS, Postman

PROFESSIONAL EXPERIENCE

Hudl India Private Limited

Mumbai, India

Wyscout Analyst

Aug 2026 – Present

- Analyze high-frequency sports telemetry and broadcast data to support advanced backend AI pipelines and predictive analytics workflows.
- Leverage proprietary rule frameworks to structure and tag kinematic spatio-temporal features, ensuring high-quality data ingestion for machine learning models.

Talent Corner HR Services Private Limited

Mumbai, India

Software Engineering Intern

Dec 2025 – Feb 2026

- Architected and optimized **PostgreSQL** database schemas to efficiently manage and query a dense relational dataset of **60,000+** detailed professional profiles, significantly minimizing search latency.
- Engineered asynchronous REST APIs using **FastAPI** and Python to power a dynamic authentication system and an automated document generation engine.
- Leveraged AI coding assistants (Cursor/Claude) to accelerate backend pipeline development, deploying via Git and conducting rigorous API load testing to ensure system stability.

AI SYSTEMS & ENGINEERING PROJECTS

Sports Analyser: Computer Vision & Vector RAG Platform | *YOLO, FastAPI, ChromaDB, OpenCV*

- Architected a real-time computer vision pipeline tracking multi-agent kinematics across broadcast footage utilizing YOLO object detection and Kalman filtering.
- Developed an active learning subsystem to automatically extract high-entropy video frames, iteratively pre-labeling them to continuously improve model confidence in production.
- Engineered a semantic vector RAG engine using ChromaDB to match real-time kinematic spatial features against 70+ proprietary business rules.

OmniMath-Local: Symbolic RAG & Reasoning SLM | *FastAPI, ChromaDB, DeepSeek, SymPy*

- Engineered an enterprise-grade async backend framework serving custom algorithmic validation logic and structured mathematical reasoning workflows.
- Implemented a local vector RAG system indexing 12,000+ technical documents, optimizing system throughput and minimizing disk boundaries via asynchronous minibatches.
- Integrated a local SLM with a symbolic verification parser to programmatically evaluate algebraic correctness, entirely eliminating mathematical hallucinations.

AURA HUD: Native Edge AI Companion | *Rust (Tauri), FastAPI, LangChain, SQLite*

- Engineered a cursor-anchored desktop AI assistant utilizing a Rust (Tauri) core to achieve a minimal memory footprint while maintaining sub-3.0s local LLM response latency.
- Architected zero-latency clipboard intelligence using FastAPI and SQLite in Write-Ahead Logging (WAL) mode to instantly parse and action copied code contexts.
- Developed a 100% offline audio streaming pipeline processing voice data over WebSockets into a local speech-to-text engine for secure transcription.

NeuroRehab-BCI: Deep Learning Signal Classifier | *PyTorch, MNE-Python, FastAPI*

- Developed a complete deep learning and signal processing pipeline to classify multi-channel EEG signals using custom temporal-spatial convolutional neural networks.
- Engineered a virtual hardware bridge to stream live data windows to a FastAPI inference server, achieving highly optimized real-time prediction latency.

EDUCATION

Nagandas Khandwala College

Mumbai, Maharashtra

B.Sc. Computer Science (Spl. in Artificial Intelligence & Machine Learning)

2023 – 2026

- **Academic Standing:** CGPA: 7.5
- **Relevant Focus:** Data Structures & Algorithms (DSA), Neural Networks, Database Management Systems (DBMS), Distributed Web Architecture.

CERTIFICATIONS

Cloud Engineering: AWS Data Engineering for Generative AI Applications, AWS Connecting Systems and Machines for Industrial Manufacturing, Microsoft Azure Cloud Computing (SDP)

Machine Learning & Data Science: DeepLearning.AI Machine Learning Specialization, TensorFlow Developer Certificate